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INFT6800 - Professional Practice in IT

Research Seminar on AI in the Professional Working Environments



Source: Bernard Marr, Forbes 2024

Executive Summary:

This research report explores the evolution, current state, and future prospects of Artificial Intelligence in professional working environments. Drawing from the McKinsey report "Superagency in the Workplace: Empowering people to unlock AI's full potential" and other academic and professional sources, we investigate how AI is transforming the workplace, its impact on the workforce, and its implications for future careers [2].

Our findings indicate that AI has rapidly evolved from theoretical concepts to powerful tools that are reshaping nearly every industry. While less than 1% of organizations describe their AI deployments as "mature," 92% plan to increase AI investments over the next three years. We found that employees are generally more ready to embrace AI than their leaders realize, with 13% of employees already using AI for at least 30% of their daily work triple what executives estimate [2].

Regarding workforce impact, our research suggests that AI will not lead to widespread job elimination but rather to job transformation. By 2030, approximately 30% of current U.S. jobs could be automated, with 60% significantly altered by AI tools [3]. However, historical patterns suggest that technological advancement creates more jobs than it eliminates over time, though requiring different skill sets.

Professional fields where AI demonstrates superior capabilities include information technology, finance, and customer sales, where 73%, 70%, and 67% of tasks respectively are expected to see significant AI impact [7]. In these fields, AI excels at tasks involving data analysis, pattern recognition, repetitive processes, and standardized decision-making.

While AI offers tremendous benefits in productivity, creativity, and problem-solving capabilities, it also presents significant challenges related to cybersecurity, accuracy, data privacy, and ethical concerns. Deepfake technology, in particular, represents a growing threat, with incidents increasing tenfold between 2022 and 2023 [10].

Looking ahead, we anticipate that successful professionals will be those who develop an "AI-plus" skill set combining domain expertise with AI literacy and uniquely human capabilities like emotional intelligence, ethical judgment, and creative problem-solving.

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1. Introduction:

1.1 Definition of Artificial Intelligence:

Artificial intelligence is a specialty within computer science concerned with creating systems that can replicate human intelligence and problem-solving abilities. Unlike conventional computer programs that require human intervention to fix bugs and improve processes, AI systems take in vast amounts of data, process it, and learn from past experiences to streamline and improve their performance over time [1].



Source: Madhu Chavva, IEEE Computer Society, 2025

The field of artificial intelligence refers to creating computer systems which perform work that needs human intelligence. The work that requires human intelligence includes visual perception alongside speech recognition and decision-making along with language translation and adaptive learning. Advanced AI systems possess the ability to examine intricate data collections and detect patterns which enables them to forecast outcomes and produce original content that was formerly accessible only to human minds.

1.2 History and Evolution of Artificial Intelligence:

The concept of artificial intelligence has ancient roots in philosophy, with thinkers contemplating the nature of life, consciousness, and artificial beings. Early mechanical automatons date back to 400 BCE, with records of a mechanical pigeon created by a friend of Plato. Leonardo da Vinci designed an automaton around 1495, demonstrating the longstanding human fascination with creating machines that mimic life [1].

The modern history of AI can be divided into several key periods:

Groundwork for AI (1900-1950):

The fundamental ideas which formed the basis of AI emerged during the first decades of the 20th century. Karel Capek first used the word "robot" during 1921 when he created his science fiction work "Rossum's Universal Robots." Computer scientist Edmund Callis Berkeley released "Giant Brains, or Machines that Think" in 1949 to describe new computer models which he compared to human brains [1].

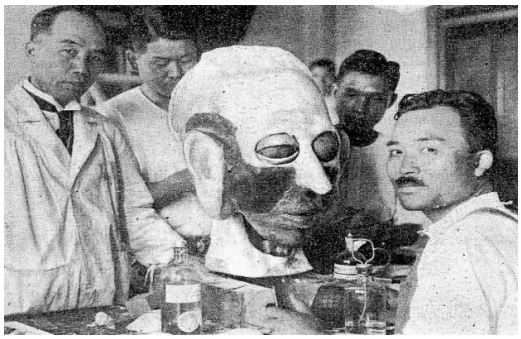


Figure: Makoto Nishimura (left) and his team designed Gakutensoku's head so that it could express human affect.
Source: Yulia Frumer, IEEE Spectrum, 2020



Karel Capek

Birth of AI (1950-1956):

This period marked the formal beginning of AI as a field. In 1950, Alan Turing published "Computer Machinery and Intelligence," which proposed the Turing Test as a measure of machine intelligence. In 1952, Arthur Samuel developed a program that could learn to play checkers independently. The term "artificial intelligence" was coined by John McCarthy in 1955 at a workshop at Dartmouth College, establishing AI as an academic discipline [1].



Arthur Lee Samuel

AI Maturation (1957-1979):

The late 1950s through the 1970s saw both rapid growth and challenges for AI research. In 1958, John McCarthy created LISP, the first programming language designed specifically for AI research. The first industrial robot, **Unimate**, began working on a General Motors assembly line in 1961. In 1966, Joseph Weizenbaum created ELIZA, the first chatbot, which used natural language processing to converse with humans. However, the 1970s also saw funding reductions for AI research due to unfulfilled promises [1].



Unimate, the grandfather of industrial robots, **Source**: Robotsguide.com

AI Boom (1980-1987):

The 1980s witnessed renewed interest and investment in AI. The first commercial expert system, XCON, entered the market in 1980. In 1986, Ernst Dickmann and his team demonstrated the first driverless car, capable of speeds up to 55 mph under controlled conditions. However, by 1987, the market for specialized AI hardware collapsed as cheaper alternatives became available [1].



Ernst DickManns, the German scientist

Source: Janosch Delcker Politico

AI Winter (1987-1993): This period saw reduced funding and interest in AI research, often referred to as the "AI Winter." Despite these challenges, development continued, with innovations like the chatbot Jabberwacky created by Rollo Carpenter in 1988 [1].

AI Agents (1993-2011): The 1990s and 2000s brought significant breakthroughs. In 1997, IBM's Deep Blue defeated world chess champion Garry Kasparov. Windows released speech recognition software in 1997. The first Roomba was introduced in 2002. Social media platforms like Facebook and Twitter began utilizing AI in their algorithms around 2006. In 2011, IBM's Watson won Jeopardy against former champions, and Apple released Siri, the first widely popular virtual assistant [1].

Artificial General Intelligence (2012-present):

Recent years have seen exponential advances in AI capabilities. In 2012, Google researchers trained a neural network to recognize cats without explicit programming. In 2016, Hanson Robotics created Sophia, a humanoid robot with realistic facial expressions. OpenAI released GPT-3 in 2020, capable of generating human-like text, and DALL-E in 2021, which can create images from text descriptions. More recently, large language models (LLMs) like GPT-4, Claude, and Gemini have demonstrated unprecedented capabilities in understanding context, generating content, and reasoning across diverse domains [1].

Department	Large Impact (%)	Small Impact (%)	No Impact (%)
IT	73	26	1
Finance	70	21	9
Customer Sales	67	16	17
Operations	65	18	17
HR	57	41	2
Marketing	56	41	3
Legal	46	50	4
Supply Chain	43	18	39

Table: Job Departments Most Impacted by AI (Source: Visual Capitalist, 2024)

1.3 Current State of AI in Professional Environments:

Today, AI has moved beyond experimental phases to become an integral part of professional environments across industries. McKinsey research sizes the long-term AI opportunity at $4.4 trillion in added productivity growth potential from corporate use cases [2]. AI systems are now deployed for various business functions, including:

- Customer service and support through chatbots and virtual assistants.
- Data analysis and business intelligence.
- Process automation and workflow optimization.
- Content creation and marketing.
- Predictive maintenance and quality control.
- Fraud detection and cybersecurity.
- Personalized recommendations and experiences.
- Healthcare diagnostics and treatment planning.
- Financial forecasting and algorithmic trading.

Despite widespread investment in AI technologies, organizational maturity remains low. According to the McKinsey report, while nearly all companies are investing in AI, only 1% of leaders describe their companies as "mature" in AI deployment, defined as having AI fully integrated into workflows and driving substantial business outcomes [2].

1.4 Future Projections:

Looking ahead, several key trends are expected to shape the future of AI in professional environments:

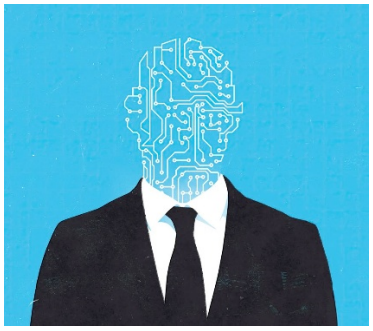
Enhanced Intelligence and Reasoning: AI systems will continue to improve in their reasoning capabilities, moving beyond information retrieval to nuanced understanding and complex decision-making.

Agentic AI: Autonomous AI agents will increasingly be able to take actions and complete complex tasks across workflows without human intervention.

Multimodality: AI will become more adept at processing and generating content across multiple modes text, audio, video, and more.

Improved Hardware: Specialized chips and increased computational power will enable faster, larger, and more versatile AI models.

Increased Transparency: AI systems will become more explainable and transparent, addressing current challenges related to bias, safety, and regulatory compliance.



Source: Jack Kelly, Forbes 2025

By 2030, AI is expected to automate approximately 30% of current jobs, with 60% significantly altered by AI tools [3]. This transformation will create new roles requiring different skill sets, particularly those combining domain expertise with AI literacy.

2. Literature Review:

This section examines recent research on AI's impact on the workforce, its applications in professional environments, and associated benefits and challenges.

2.1 AI and Workforce Transformation:

Research consistently shows that AI is transforming the nature of work rather than simply eliminating jobs. According to Frank et al. (2019) in "Toward understanding the impact of

artificial intelligence on labor," AI's impact on employment should be analysed at the task level rather than the job level, as most occupations consist of multiple tasks, only some of which are suitable for automation. Their research suggests that jobs will be redefined rather than eliminated wholesale [4].

McKinsey's "Superagency in the Workplace" report 2025 finds that employees are more ready for AI than their leaders imagine. The study reveals that 13% of employees currently use AI for at least 30% of their daily work triple what C-suite executives estimate 4%. Furthermore, 47% of employees believe AI will handle more than 30% of their daily tasks within a year, compared to only 20% of leaders who share this expectation [2].

Hazarika 2020 examines the impact of AI on the healthcare workforce, finding that while AI can automate routine tasks, it also creates new opportunities for healthcare professionals to focus on more complex and interpersonally demanding aspects of patient care. This pattern of task redistribution, rather than wholesale job elimination, is consistent across multiple professional fields [5].

2.2 Professional Applications of AI:

Wisskirchen et al. 2017 provide a comprehensive analysis of AI and robotics in the workplace, identifying key professional domains where AI is making significant inroads. Their research highlights legal, ethical, and social implications of AI adoption across industries and emphasizes the need for regulatory frameworks that balance innovation with worker protection [6].

A 2024 World Economic Forum report, visualized by Visual Capitalist, identifies the professional departments most impacted by AI. Information technology 73%, finance 70%, and customer sales 67% show the highest percentages of tasks likely to be significantly impacted by AI technologies. The report assessed over 19,000 occupational tasks to determine their reliance on language and the degree of human involvement required [7].

Morandini et al. 2023 examine the transformation of professional skills by AI, focusing on upskilling and reskilling in organizations. Their research identifies a shift from technical skills to higher-order cognitive and social skills that complement AI capabilities. They argue that organizations must invest in continuous learning programs to prepare their workforce for collaboration with AI systems [8].

2.3 Benefits and Challenges

Tredinnick 2017 explores AI's impact on professional roles, highlighting both the opportunities for enhanced productivity and the anxieties surrounding job displacement. The research emphasizes that successful AI integration requires careful consideration of human-machine collaboration rather than simple substitution [9].

Security.org's 2024 deepfake statistics report highlights the growing threats posed by AI-generated content. Deepfake fraud incidents increased tenfold between 2022 and 2023, with 88% of identified cases occurring in the cryptocurrency sector. The report notes that 70% of

people are not confident in their ability to distinguish between real and cloned voices, creating significant vulnerabilities for social engineering attacks [10].

The Stanford University Center for Research on Foundation Models reports on transparency in AI systems, tracking improvements through their Transparency Index. While progress has been made with Anthropic's transparency score increasing by 15 points to 51 and Amazon's more than tripling to 41 between October 2023 and May 2024 significant challenges remain in making AI systems fully explainable and accountable [2].

3. Problems and Discussions:

3.1 Will AI Replace People and Reduce the Human Workforce:

The question of whether AI will replace human workers has been the subject of extensive research and debate. Our analysis suggests a more subtle reality than wholesale replacement.

3.1.1 Statistical Evidence on Job Transformation:

Current research indicates that AI will transform jobs rather than eliminate the workforce entirely:



- By 2030, approximately 30% of current jobs could be automated, with 60% significantly altered by AI tools [3].
- According to McKinsey, by 2030, 14% of the global workforce roughly 375 million workers may need to change careers due to AI and automation [11].
- A MIT and Boston University report suggests AI will replace as many as two million manufacturing workers by 2025 [3].
- Generative AI could disrupt 50% of the tasks performed by market research analysts (53%) and sales representatives (67%) [13].

3.1.2 Historical Patterns of Technological Disruption

Historical evidence suggests that while technological advancements disrupt existing jobs, they typically create more new jobs than they eliminate:

- The Industrial Revolution eliminated many manual crafting jobs but created millions of new manufacturing positions.
- The advent of ATMs in the banking industry initially reduced the need for tellers per branch but led to an overall increase in bank branches and total employment in the sector.
- The rise of the internet eliminated some traditional roles but created entirely new industries and job categories that didn't previously exist.

This pattern suggests that AI will likely follow a similar trajectory eliminating or transforming some jobs while creating new ones that we cannot yet fully envision.

3.1.3 Factors Influencing Job Displacement vs. Creation

Several factors will determine whether AI leads to net job displacement or creation:

1. **Task Composition:** Jobs composed primarily of routine, predictable tasks are more susceptible to automation than those requiring complex problem-solving, creativity, or emotional intelligence.
2. **Economic Dynamics:** Productivity gains from AI may lower costs and increase demand for products and services, potentially creating new jobs to meet this demand.
3. **Skill Adaptation:** The workforce's ability to develop new skills that complement AI capabilities will influence employment outcomes.
4. **Policy Responses:** Government policies regarding education, training, and social safety nets will play a crucial role in managing the transition.
5. **Industry Differences:** Some industries will see more job creation than others. For example, healthcare may see significant job growth as AI enables more personalized care models.

3.1.4 Analysis and Discussion:

Based on our research, we conclude that AI will not render humans redundant but will require significant adaptation. The concept of "superagency" introduced in the McKinsey report offers a useful framework AI as a force multiplier for human agency rather than a replacement for it [2].

Evidence suggests that workers who learn to collaborate effectively with AI tools will be more productive and valuable than either AI systems or human workers operating independently. This "AI-plus" skill set combining domain expertise with AI literacy will likely define successful professionals in the coming decades.

Furthermore, certain uniquely human capabilities remain difficult or impossible to automate:

- Emotional intelligence and empathy.
- Ethical judgment and moral reasoning.

- Creative problem-solving in novel situations.
- Building trust and relationships.
- Cultural sensitivity and contextual understanding.

These capabilities suggest that human workers will remain essential, albeit in roles that may look different from today's jobs. The workforce will not be reduced but rather redistributed and reskilled to perform higher-value tasks that complement AI capabilities.

3.2 Professional Jobs Where AI Performs Better:

Current research indicates that AI excels at certain types of professional tasks and is increasingly outperforming humans in specific domains. This section examines which professional jobs or tasks AI performs better at and analyses their cost-effectiveness.

3.2.1 Task Categories Where AI Excels:

AI systems demonstrate superior performance in tasks with the following characteristics:

Data-Intensive Analysis: Tasks requiring processing and analysis of vast quantities of data, where human cognitive limitations become a bottleneck.

Pattern Recognition: Identifying subtle patterns across large datasets that might be imperceptible to human analysts.

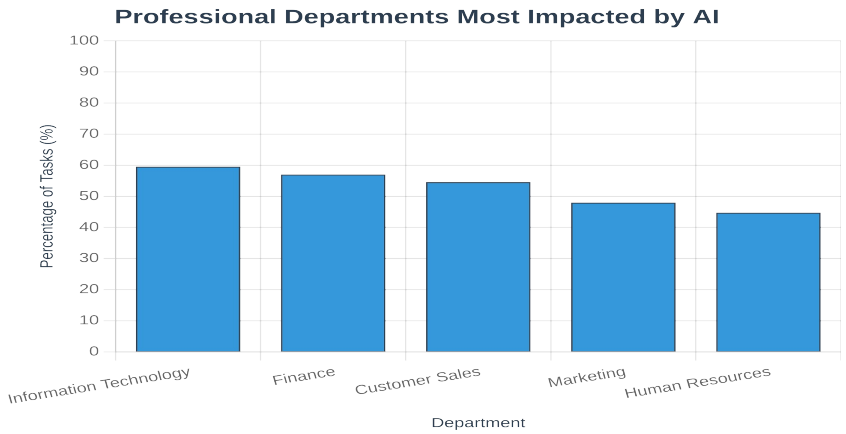
Repetitive Processes: Tasks that follow consistent rules and patterns with minimal variation.

Rapid Information Retrieval: Quickly accessing and synthesizing information from large knowledge bases.

Standardized Decision-Making: Making decisions based on well-defined criteria and substantial historical data.

3.2.2 Professional Fields with High AI Impact

According to the World Economic Forum report visualized by Visual Capitalist 2024, the professional departments most impacted by AI include [7]:



Source: Visual Capitalist, 2024

Within these departments, specific professional roles where AI is demonstrating superior performance include:

Information Technology:

Software Quality Assurance: AI can test software more comprehensively and efficiently than human testers, identifying bugs and vulnerabilities across countless scenarios.

IT Support: AI chatbots can resolve common technical issues faster than human agents, with 24/7 availability and consistent quality.

Code Generation: AI coding assistants can generate boilerplate code and suggest optimizations more rapidly than human programmers.

Finance:

Bookkeeping and Accounting: AI systems can process transactions, reconcile accounts, and prepare financial statements with greater accuracy and speed than human accountants.

Fraud Detection: AI algorithms can identify suspicious patterns across millions of transactions in real-time, outperforming human analysts in both speed and accuracy.

Investment Analysis: AI can analyse market trends, company performance, and economic indicators simultaneously, identifying investment opportunities that human analysts might miss.

Customer Sales and Service:

Customer Inquiry Handling: AI chatbots can handle routine customer inquiries consistently and at scale, without fatigue or emotional variability.

Sales Lead Qualification: AI systems can analyse prospect data and behaviors to identify high-potential leads more accurately than human sales representatives.

Product Recommendations: AI can process customer preferences and purchase history to generate personalized recommendations more effectively than human sales associates.

3.2.3 Cost-Effectiveness Analysis

The cost-effectiveness of AI compared to human workers varies by context but shows clear advantages in many scenarios:

Scale Efficiency: Once developed, AI systems can be deployed at scale with minimal marginal cost, unlike human workers whose costs scale linearly with workload.

24/7 Availability: AI systems can operate continuously without overtime costs, sick leave, or vacation time.

Consistent Quality: AI delivers consistent performance without fatigue, emotional variation, or human error.

Implementation Costs: Initial development and implementation costs for AI systems can be substantial but typically amortize over time and scale.

McKinsey research indicates that while only 19% of companies report revenue increases exceeding 5% from AI adoption, the ROI improves significantly as organizations mature in their AI capabilities [2]. Companies at more advanced stages of AI maturity report both revenue growth and cost reduction at higher rates than those at earlier stages.

3.2.4 Analysis and Discussion:

Our analysis suggests that AI is increasingly capable of outperforming humans in specific professional contexts, particularly those involving data processing, pattern recognition, and rule-based decision-making. However, this superiority is highly task-specific rather than job-specific.

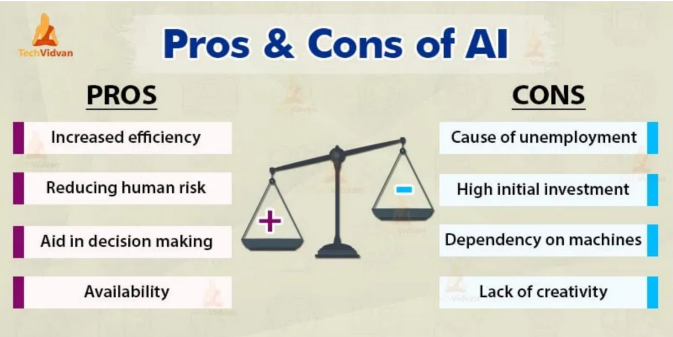
Most professional roles comprise multiple tasks, some well-suited to AI automation and others requiring uniquely human capabilities. The most effective approach appears to be human-AI collaboration, where AI handles routine, data-intensive tasks while humans focus on judgment, creativity, and interpersonal aspects.

The cost-effectiveness of AI solutions improves with scale and over time, suggesting that larger organizations and those with longer implementation horizons will realize greater benefits. However, the full cost calculation must include not only technology implementation

but also organizational change management, training, and potential disruption during transition periods.

3.3 Positives and Negatives of AI:

AI technologies offer significant benefits while also presenting substantial challenges. This section examines the positives and negatives of AI, with particular attention to issues such as deepfakes, plagiarism, and security concerns.



Source: Tredinnick (2017), McKinsey report "Superagency in the Workplace" (2025)

3.3.1 Positive Impacts of AI

Productivity Enhancement:

AI significantly boosts productivity across professional environments by:

Automating Routine Tasks: Freeing human workers to focus on higher-value activities.

Accelerating Information Processing: Analysing vast datasets faster than humanly possible.

Reducing Error Rates: Minimizing mistakes in repetitive, detail-oriented tasks.

Enabling 24/7 Operations: Providing continuous service without fatigue.

McKinsey research estimates the long-term AI opportunity at $4.4 trillion in added productivity growth potential from corporate use cases [2].

Enhanced Decision-Making:

AI improves decision-making processes by:

Identifying Patterns: Recognizing trends and correlations in complex data.

Reducing Cognitive Biases: Providing data-driven insights less influenced by human biases.

Enabling Predictive Analytics: Forecasting outcomes based on historical data.

Supporting Evidence-Based Approaches: Bringing more information to bear on decisions.

Democratization of Expertise:

AI helps democratize access to specialized knowledge by:

Lowering Skill Barriers: Enabling non-experts to perform complex tasks with AI assistance.

Expanding Access to Information: Making specialized knowledge more widely available.

Enabling Self-Service: Allowing users to address needs without specialist intervention.

Supporting Education: Providing personalized learning experiences at scale.

Innovation Acceleration:

AI accelerates innovation by:

Augmenting Human Creativity: Generating novel ideas and approaches.

Simulating Outcomes: Testing hypotheses without physical prototyping.

Cross-Domain Inspiration: Identifying connections between disparate fields.

Rapid Prototyping: Accelerating the iteration process.

3.3.2 Negative Impacts of AI:



Deepfake Concerns:

Deepfakes AI-generated media that convincingly depict events that never occurred present growing challenges:

Rapid Proliferation: Deepfake fraud incidents increased tenfold between 2022 and 2023 [10].

Detection Difficulty: 70% of people report being unable to confidently distinguish between real and cloned voices [10].

Credential Bypass: Voice deepfakes have successfully bypassed voice authentication systems at financial institutions

Financial Impact: CEO fraud targets at least 400 companies per day, with damages reaching up to 10% of annual profits for successful attacks [10].

Organizational Unpreparedness: 80% of companies lack protocols to handle deepfake attacks [10].

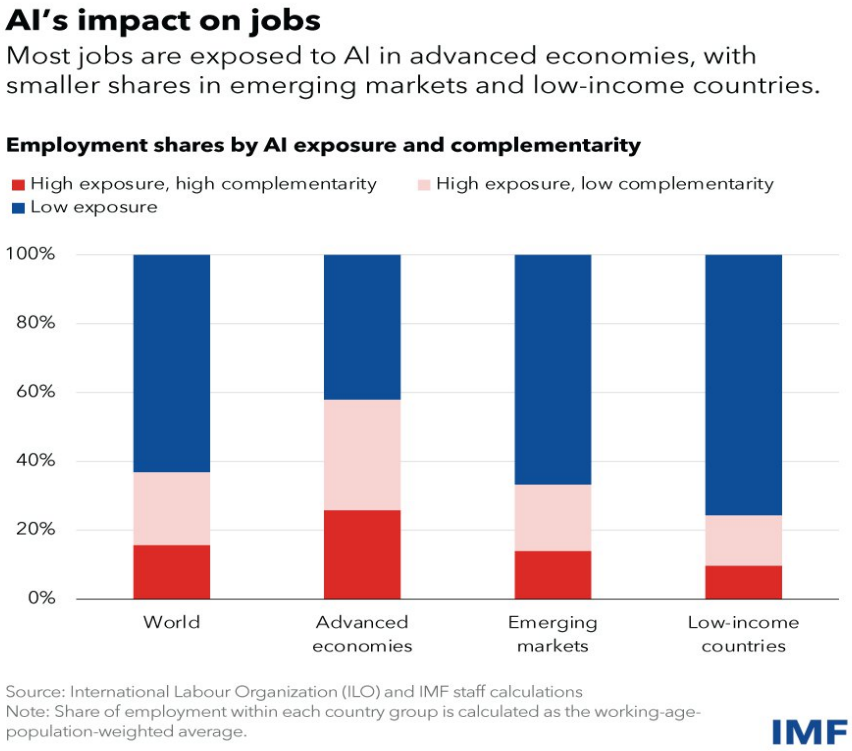


Figure: AI Impact on Jobs (Source: Weam AI, 2024)

Security and Privacy Vulnerabilities

AI creates new security and privacy concerns including:

Data Exposure: AI systems require vast amounts of data, creating potential privacy vulnerabilities.

Adversarial Attacks: Malicious inputs designed to manipulate AI outputs.

Model Inversion: Techniques that can extract training data from AI models.

Enhanced Phishing: AI-generated phishing attacks that are more convincing and personalized.

Cybersecurity Arms Race: Both defensive and offensive security increasingly relying on AI.

According to employee surveys, cybersecurity risks 51% and concerns about personal privacy 43% rank among the top concerns regarding AI adoption in the workplace [2].

Accuracy and Reliability Issues:

AI systems face challenges related to accuracy and reliability:

Hallucinations: AI generating plausible but factually incorrect information.

Bias Amplification: AI systems perpetuating or amplifying biases present in training data.

Context Misunderstanding: Failure to grasp nuanced contextual factors.

Overconfidence: Presenting uncertain conclusions with unwarranted certainty.

Unpredictable Failures: Unexpected errors that may be difficult to diagnose.

Employee surveys indicate that inaccuracies 50% are a major concern regarding AI adoption [2].

Intellectual Property and Plagiarism:

AI raises complex questions about intellectual property and originality:

Content Derivation: AI-generated content derived from copyrighted materials.

Attribution Challenges: Difficulty in determining the sources that influenced AI outputs.

Automated Plagiarism: AI systems reproducing content without proper attribution.

Copyright Uncertainty: Evolving legal frameworks struggling to address AI-generated works.

Academic Integrity: Challenges in distinguishing between human and AI-authored content.

The U.S. Copyright Office currently considers AI-generated works on a case-by-case basis and welcomes public input on these issues [10].

3.3.3 Analysis and Discussion:

The rapid advancement of AI technologies presents a complex balance of benefits and risks. While the productivity, decision-making, and innovation benefits are substantial, the challenges related to deepfakes, security, accuracy, and intellectual property cannot be overlooked.

Addressing these challenges requires a multi-faceted approach:

Technical Solutions: Developing better detection systems for deepfakes, improving AI model transparency, and implementing robust security measures.

Regulatory Frameworks: Establishing clear guidelines for AI use, particularly in sensitive contexts.

Organizational Policies: Implementing protocols for AI governance, including benchmarking and monitoring.

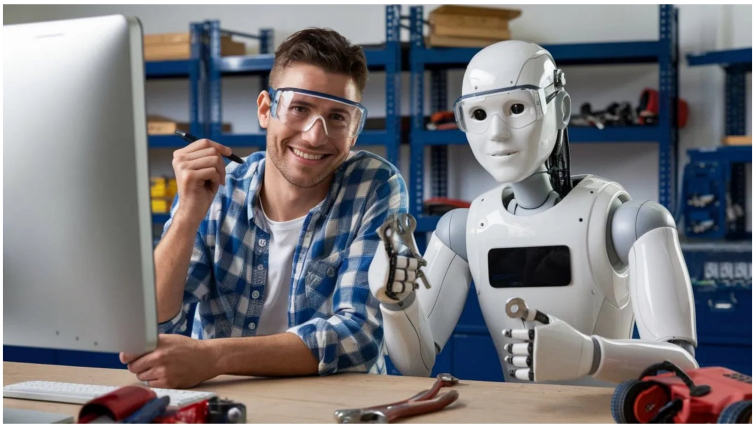
Education and Training: Ensuring users understand both AI capabilities and limitations.

Ethical Standards: Developing industry-wide ethical principles for responsible AI development and use.

Notably, employees place high trust in their employers to deploy AI responsibly 71% trust their employers to act ethically as they develop AI, higher than their trust in universities 67%, large tech companies 61%, or startups 51% [2]. This trust places significant responsibility on organizational leaders to implement AI in ways that mitigate risks while capturing benefits.

4. Reflection Discussion:

As a group reflecting on our future professional careers, we considered how AI might impact our chosen fields and whether we feel threatened by potential replacement over time. Our reflection encompasses multiple perspectives across different professional domains.



4.1 Information Technology Professional

As aspiring IT professionals, we recognize that our field shows the highest potential for AI impact, with 73% of tasks susceptible to significant disruption according to the World

Economic Forum report [7]. However, rather than feeling threatened, we see this as an opportunity for evolution.

While AI can now generate code, detect bugs, and automate testing more efficiently than humans, the role of IT professionals is shifting toward higher-value activities:

- Designing AI systems and determining appropriate use cases.
- Managing the integration of AI into existing systems and workflows.
- Ensuring security, privacy, and ethical compliance of AI deployments.
- Interpreting and explaining AI outputs to non-technical stakeholders.
- Addressing complex, novel problems that require creative solutions.

Rather than replacing IT professionals, AI is becoming our most powerful tool, amplifying our capabilities and allowing us to focus on more creative and strategic aspects of technology development and implementation. The concept of "superagency" described in the McKinsey report resonates strongly AI as a force multiplier for human agency rather than a replacement for it [2].

4.2 Finance Professional

For those of us interested in finance careers, the high susceptibility of financial tasks to AI automation 70% according to the WEF report initially appears concerning [7]. However, examining the nature of financial work reveals a more nuanced picture.

While AI excels at data processing, transaction analysis, and pattern recognition automating tasks like bookkeeping, fraud detection, and basic financial analysis the core value of finance professionals increasingly lies in:

- Strategic financial planning that considers organizational context and goals.
- Building client relationships based on trust and understanding of personal circumstances.
- Making judgment calls in complex or ambiguous situations.
- Providing ethical guidance on financial decisions.
- Translating financial insights into strategic business decisions.

We anticipate that finance roles will evolve to place greater emphasis on these human-centric aspects, with AI handling the computational and routine analytical components. This evolution aligns with the historical pattern of technological advancement in finance from calculators to spreadsheets to algorithmic trading each innovation shifting the role of finance professionals rather than eliminating it.

4.3 Marketing Professional

In marketing, AI's impact 56% of tasks likely to see significant impact according to the WEF report is already evident in content generation, campaign optimization, and customer segmentation [7]. As aspiring marketing professionals, we see this transformation as both challenging and promising.

While AI can generate content, analyse customer data, and optimize campaign performance more efficiently than humans, the essence of effective marketing remains deeply human:

- Understanding emotional resonance and cultural nuance.
- Developing authentic brand voices that connect with audiences.
- Creating innovative campaign concepts that capture attention.
- Building relationships with partners, influencers, and customers.
- Interpreting market trends in the context of broader social changes.

We envision future marketing roles leveraging AI as a collaborative partner generating initial content drafts, identifying optimization opportunities, and managing routine communications, while human marketers focus on strategy, creativity, and relationship building. This collaboration model aligns with the findings in the McKinsey report that millennials 35-44 age group are the most comfortable using AI at work 90% and providing feedback on AI tools [2].

4.4 Data Science Professional:

For those pursuing careers in data science, the relationship with AI is particularly complex, as we are both creators and potential subjects of automation. Advanced AI systems can now perform many data science tasks, from data cleaning to feature engineering to model selection, with increasing autonomy.

However, we believe the role of data scientists will evolve rather than disappear:

- Shifting from implementing standard models to designing novel approaches for complex problems.
- Moving from data processing to problem formulation and interpretation.
- Focusing more on ethical considerations and social implications of data use.
- Serving as translators between technical capabilities and business needs.
- Developing expertise in AI governance and responsible deployment.

The democratization of basic data science capabilities through AI tools means that data scientists will need to develop deeper expertise and focus on more complex problem-solving

to remain valuable. This aligns with the historical pattern of technological advancement as basic capabilities become accessible to more people, specialists move up the value chain to more sophisticated applications.

4.5 Synthesis and Conclusion:

Across our diverse professional interests, we see a consistent pattern: AI will not replace us in our chosen roles, but it will transform how we work and what we focus on. The key to thriving in this transformed landscape will be developing an "AI-plus" skill set that combines:

Domain Expertise: Deep understanding of our respective fields.

AI Literacy: The ability to work effectively with AI tools.

Human-Centric Capabilities: Skills like creativity, empathy, ethical judgment, and relationship building that remain distinctively human.

Adaptability: The willingness to continuously learn and evolve our skills.

Systems Thinking: The ability to understand complex interactions between technology, people, and processes.

Rather than feeling threatened by AI, we are excited about the potential for "superagency" using AI to amplify our capabilities and impact. As the McKinsey report suggests, "AI could drive enormous positive and disruptive change" [2]. Our challenge and opportunity will be to navigate this change proactively, shaping how AI is integrated into our chosen professions rather than simply reacting to its advancement.

5. Conclusion:

This research has examined the evolution, current state, and future prospects of AI in professional working environments. Through our analysis of the McKinsey "Superagency in the Workplace" report and additional academic and professional sources, we have gained insights into how AI is transforming the nature of work across industries [2].

Our investigation of AI's historical development reveals a trajectory from theoretical concepts to increasingly capable systems that can now reason, generate content, and make autonomous decisions. While less than 1% of organizations describe their AI deployments as "mature," the technology continues to advance rapidly, with 92% of companies planning to increase their AI investments over the next three years [2].

Regarding AI's impact on the workforce, our research suggests that fears of wholesale human replacement are largely unfounded. Instead, we observe a pattern of job transformation, with AI automating routine tasks while creating new opportunities that leverage uniquely human capabilities. By 2030, approximately 30% of current U.S. jobs could be automated, with 60% significantly altered by AI tools but historical patterns suggest that technological advancement typically creates more jobs than it eliminates over time [3].

Our analysis of professional domains reveals that information technology 73%, finance 70%, and customer sales 67% show the highest percentages of tasks likely to be significantly impacted by AI [7]. In these fields, AI excels at tasks involving data analysis, pattern recognition, repetitive processes, and standardized decision-making. However, the cost-effectiveness of AI solutions varies by context and improves with scale and over time.

We identified significant benefits of AI in professional environments, including productivity enhancement, improved decision-making, democratization of expertise, and innovation acceleration. However, these benefits come with substantial challenges related to deepfakes which increased tenfold between 2022 and 2023, security vulnerabilities, accuracy issues, and intellectual property concerns [10].

Our reflection on future careers across various professional domains suggests that AI will transform rather than eliminate our chosen fields. Success will depend on developing an "AI-plus" skill set that combines domain expertise, AI literacy, and uniquely human capabilities like creativity, empathy, and ethical judgment.

Perhaps most encouragingly, our research found that employees are generally more ready to embrace AI than their leaders realize. They place high trust in their employers to deploy AI responsibly 71% trust their employers to act ethically as they develop AI, higher than their trust in other institutions [2]. This trust provides a foundation for responsible AI adoption that balances innovation with ethical considerations.

The concept of "superagency" introduced in the McKinsey report offers a compelling vision for the future of work AI as a force multiplier for human agency rather than a replacement for it [2]. By embracing this vision and proactively developing the skills needed to collaborate effectively with AI systems, professionals across all fields can thrive in the transformed workplace of tomorrow.

As we conclude this research, we recognize that AI's evolution continues at a rapid pace. The integration of AI into professional environments will require ongoing adaptation, learning, and thoughtful governance. By understanding both the capabilities and limitations of AI technologies, organizations and individuals can navigate this transformation to realize the full potential of human-AI collaboration.

References:

[1] Tableau. (2024). What is the history of artificial intelligence (AI)? Retrieved from https://www.tableau.com/data-insights/ai/history

[2] McKinsey. (2025). Superagency in the Workplace: Empowering people to unlock AI's full potential. Retrieved from https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/superagency-in-the-workplace-empowering-people-to-unlock-ais-full-potential-at-work

[3] Forbes. (2025). Jobs AI Will Replace First in the Workplace Shift. Retrieved from https://www.forbes.com/sites/jackkelly/2025/04/25/the-jobs-that-will-fall-first-as-ai-takes-over-the-workplace/

[4] Frank, M. R., Autor, D., Bessen, J. E., Brynjolfsson, E., et al. (2019). Toward understanding the impact of artificial intelligence on labor. Proceedings of the National Academy of Sciences. Retrieved from https://www.pnas.org/doi/abs/10.1073/pnas.1900949116

[5] Hazarika, I. (2020). Artificial intelligence: opportunities and implications for the health workforce. International Health. Retrieved from https://academic.oup.com/inthealth/article-pdf/doi/10.1093/inthealth/ihaa007/33441053/ihaa007.pdf

[6] Wisskirchen, G., Biacabe, B. T., Bormann, U., et al. (2017). Artificial intelligence and robotics and their impact on the workplace. IBA Global Employment Institute. Retrieved from https://www.researchgate.net/profile/Mohamed-Mourad-Lafifi/post/Social_Robots_or_robots_with_social_functions/attachment/6001ed617e98b40001bc005a/AS%3A980324746031116%401610739041600/download/AI-and-Robotics-IBA-GEI-April-2017.pdf

[7] Visual Capitalist. (2024). Chart: The Jobs Most Impacted by AI. Retrieved from https://www.visualcapitalist.com/charted-the-jobs-most-impacted-by-ai/

[8] Morandini, S., Fraboni, F., De Angelis, M., Puzzo, G., et al. (2023). The impact of artificial intelligence on workers' skills: Upskilling and reskilling in organisations. Informing Science. Retrieved from https://cris.unibo.it/bitstream/11585/917132/1/InfoSciV26p039-068Morandini8895.pdf

[9] Tredinnick, L. (2017). Artificial intelligence and professional roles. Business Information Review. Retrieved from https://journals.sagepub.com/doi/abs/10.1177/0266382117692621

[10] Security.org. (2024). 2024 Deepfakes Guide and Statistics. Retrieved from https://www.security.org/resources/deepfake-statistics/

[11] Exploding Topics. (2025). 60+ Stats On AI Replacing Jobs (2025). Retrieved from https://explodingtopics.com/blog/ai-replacing-jobs

[12] Weam AI. (2024). AI Jobs Statistics That Will Shock You in 2024. Retrieved from https://weam.ai/blog/guide/ai-jobs-statistics/

[13] World Economic Forum. (2025). Jobs of Tomorrow: Large Language Models and Jobs. Retrieved from https://www.weforum.org/publications/jobs-of-tomorrow-large-language-models-and-jobs/

[14] Team, R. (2023, December 19). *Unimate*. ROBOTS: Your Guide to the World of Robotics. https://robotsguide.com/robots/unimate/

[15] Marr, B. (2024, August 16). *Where will artificial intelligence take us in the future?* Forbes. https://www.forbes.com/sites/bernardmarr/2024/08/16/where-will-artificial-intelligence-take-us-in-the-future/

[16] Delcker, J. (2019, April 18). The man who invented the self-driving car (in 1986). *POLITICO*. https://www.politico.eu/article/delf-driving-car-born-1986-ernst-dickmanns-mercedes/